



MISO DPP 2018 April Michigan Area Study Phase 1 Report

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(Final Report-Version 2)

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Contents

1. Executive Summary	4
1.1. Project List.....	4
1.2. Total Network Upgrades	6
2. FERC Order 827 Compliance Review.....	9
3. Model Development and Study Assumptions.....	10
3.1. Base Case Models	10
3.2. Monitored Elements	10
3.3. Contingencies.....	10
3.4. Study Methodology	11
3.5. Performance Criteria	11
4. Thermal Analysis	11
4.1. J978.....	11
4.2. J1043.....	12
4.3. J1064.....	12
4.4. J1065.....	12
4.5. J1071.....	12
4.6. J1088.....	12
4.7. J1089.....	13
4.8. J1119.....	13
4.9. J1173.....	13
5. Voltage Analysis	14
6. Deliverability Analysis	14
6.1. Introduction.....	14
6.2. Determining the MW Restriction	14
6.3. Deliverability Study Results.....	14
6.4. J978.....	14
6.5. J984.....	14
6.6. J989.....	14
6.7. J996.....	15
6.8. J1005.....	15
6.9. J1013.....	15
6.10. J1014.....	15
6.11. J1043.....	15
6.12. J1046.....	15
6.13. J1056.....	15
6.14. J1062.....	15
6.15. J1064.....	15
6.16. J1065.....	15

6.17. J1068.....	15
6.18. J1071.....	15
6.19. J1083.....	16
6.20. J1088.....	16
6.21. J1089.....	16
6.22. J1090.....	16
6.23. J1103.....	16
6.24. J1117.....	16
6.25. J1119.....	16
6.26. J1172.....	17
6.27. J1173.....	17
6.28. J1178.....	17
7. Shared Network Upgrades Analysis	17
8. Cost Allocation	17
8.1. Cost Assumptions for Network Upgrades	17
8.2. Cost Allocation Methodology.....	17
Appendix A – Cost Allocation Summary (CEII).....	19
Appendix B – Network Upgrade Table Per Project (CEII)	19
Appendix C – MISO ERIS Analysis (CEII)	19
Appendix D – MISO Deliverability Analysis (CEII)	19
 Table 1: List of DPP 2018 April Michigan Area Phase 1 Projects	 4
Table 2: Total Cost of Network Upgrades for DPP 2018 April Michigan Phase 1 Projects.....	6
Table 3: ERIS & NRIS Upgrades (Planning level cost estimates).....	7
Table 4: Shared Network Upgrades (Planning level cost estimates)	8
Table 5: FERC Order 827 Review Results	9
Table 6: DPP 2018 April Michigan-Scenarios.....	10
Table 7: NRIS Results for J989	14
Table 8: NRIS Results for J1065	15
Table 9: NRIS Results for J1088	16
Table 10: NRIS Results for J1089	16
Table 11: Maximum MW Impact and SNU Cost Allocations	17



1. Executive Summary

This report presents the results of a System Impact Study (SIS) performed to evaluate the interconnection of the generators in the DPP 2018 April Michigan Area Phase 1 (Michigan Area DPP-2018-April Phase 1). The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies:

- ATC
- MPower
- Consumers Energy
- PJM
- NIPSCO
- DTE
- Wolverine
- ITC

1.1. Project List

The original interconnection requests for DPP 2018 April Michigan Area had a total of 36 generation projects. Projects J960, J969, J970, J1004, J1015, J1048, J1052, J1075, J1077, J1126, and J1141 withdrew prior to the start of Phase 1. Therefore, there are 25 projects generation projects with a combined nameplate rating of 3,719.4 MW (ERIS) & 3,719.4 MW (NRIS). The detailed list of Michigan Area DPP 1 is shown below in Table 1, and the Phase 1 study was kicked off on May 1 2019.

Table 1: List of DPP 2018 April Michigan Area Phase 1 Projects

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	POI
J978	Wind	METC	Montcalm County	Michigan	NRIS	100	Eureka - Vestaburg 138kV Line Tap
J984	Wind	METC	Gratiot County	Michigan	NRIS	200	Nelson Rd. 345 kV Substation
J989	Solar	METC	Oakland County	Michigan	NRIS	80	Halsey 138kV Substation
J996	Solar	METC	Lenawee County	Michigan	NRIS	80	Beecher - Moraco 138kV Line
J1005	Solar	METC	Midland County,Saginaw County	Michigan	NRIS	200	Murphy 345kV Substation
J1013	Solar	METC	Branch County	Michigan	NRIS	80	Batavia 138kV Substation
J1014	Solar	METC	Jackson County	Michigan	NRIS	80	Tompkins - Blackman - Blackstone 138kV Line
J1043	Wind	METC	Montcalm County	Michigan	NRIS	374.4	Vergennes - Nelson Rd 345 kV Line Tap
J1046	Battery Storage	ITC Transmission	Washtenaw County	Michigan	NRIS	50	Majestic - Lemoyne 345 kV Line Tap

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	POI
J1056	Wind	ITC Transmission	Monroe County	Michigan	NRIS	200	Lulu - Monroe 345 kV Line Tap
J1062	Solar	ITC Transmission	Washtenaw County	Michigan	NRIS	200	Majestic - Lemoyne 345 kV Line Tap
J1064	Wind	METC	Ionia County	Michigan	NRIS	200	Marquette - Vergennes 138 kV Line Tap
J1065	Wind	METC	Gratiot County	Michigan	NRIS	200	Alma - Vestaburg 138kV Line Tap
J1068	Wind	ITC Transmission	Washtenaw County	Michigan	NRIS	200	Dorset 120 kV Substation
J1071	Solar	METC	Oceana County	Michigan	NRIS	100	Donaldson 138 kV Substation
J1083	Solar	METC	Eaton County	Michigan	NRIS	70	Oneida 138kV Substation
J1088	Solar	METC	Shiawassee County	Michigan	NRIS	150	Cornell - Layton 138 kV Line Tap
J1089	Solar	METC	Shiawassee County	Michigan	NRIS	170	Cornell - Bell Rd 138 kV Line Tap
J1090	Solar	METC	Ingham County	Michigan	NRIS	90	Tompkins - Churchill Jct 138 kV Line Tap
J1103	Battery Storage	ITC Transmission	Tuscola County	Michigan	NRIS	20	Bauer - (Kirk) - Ringle 345kV Line Tap
J1117	Solar	METC	Ingham County	Michigan	NRIS	80	Green 138kV Substation
J1119	Gas	METC	Jackson County	Michigan	NRIS	600	Tompkins 345 kV Substation
J1172	Solar	METC	Genesee County	Michigan	NRIS	50	Dort - Garfield 138 kV Line Tap
J1173	Solar	METC	Lenawee County	Michigan	NRIS	80	Raisin - METC Tap 138kV Line Tap
J1178	Solar	METC	Eaton County	Michigan	NRIS	65	Oneida 138 kV Substation



1.2. Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2018 April Michigan Phase 1 reflects responsibilities for mitigating system impacts. The total cost of network upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies. Details pertaining to the cost allocation has been detailed in [Appendix A – Cost Allocation Summary \(CEII\)](#)

Table 2: Total Cost of Network Upgrades for DPP 2018 April Michigan Phase 1 Projects

Project	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)
	Thermal	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned				
a	b	c	d	e	f	g	h	i	j = b+c+d+f+g+i	\$10,000/MW	10% of (j) from Phase I-M2
J978	\$8,566,066	\$-	\$-	\$-	\$-	\$10,200,000	\$500,000	\$720,023	\$19,486,090	\$400,000	\$1,548,609
J984	\$-	\$-	\$-	\$-	\$-	\$2,100,000	\$500,000	\$-	\$2,100,000	\$800,000	\$-
J989	\$-	\$-	\$-	\$-	\$-	\$1,900,000	\$500,000	\$-	\$1,900,000	\$320,000	\$-
J996	\$-	\$-	\$-	\$-	\$-	\$5,500,000	\$500,000	\$-	\$5,500,000	\$320,000	\$230,000
J1005	\$-	\$-	\$-	\$-	\$-	\$2,100,000	\$500,000	\$-	\$2,100,000	\$800,000	\$-
J1013	\$-	\$-	\$-	\$-	\$-	\$1,100,000	\$500,000	\$-	\$1,100,000	\$320,000	\$-
J1014	\$-	\$-	\$-	\$-	\$-	\$7,700,000	\$500,000	\$-	\$7,700,000	\$320,000	\$450,000
J1043	\$-	\$-	\$-	\$-	\$-	\$10,700,000	\$500,000	\$-	\$10,700,000	\$1,500,000	\$-
J1046	\$-	\$-	\$-	\$-	\$-	\$6,150,000	\$250,000	\$-	\$6,150,000	\$200,000	\$415,000
J1056	\$-	\$-	\$-	\$-	\$-	\$11,400,000	\$500,000	\$-	\$11,400,000	\$800,000	\$340,000
J1062	\$-	\$-	\$-	\$-	\$-	\$6,150,000	\$250,000	\$-	\$6,150,000	\$800,000	\$-
J1064	\$-	\$-	\$-	\$-	\$-	\$8,500,000	\$500,000	\$-	\$8,500,000	\$800,000	\$50,000
J1065	\$17,644,934	\$-	\$-	\$-	\$18,000,000	\$6,900,000	\$500,000	\$1,372,797	\$43,917,730	\$800,000	\$3,591,773
J1068	\$-	\$-	\$-	\$-	\$-	\$500,000	\$1,000,000	\$-	\$500,000	\$800,000	\$-
J1071	\$19,420,000	\$-	\$-	\$-	\$-	\$1,100,000	\$500,000	\$-	\$20,520,000	\$400,000	\$1,652,000
J1083	\$-	\$-	\$-	\$-	\$-	\$750,000	\$500,000	\$-	\$750,000	\$280,000	\$-
J1088	\$331,281	\$-	\$-	\$-	\$2,736,931	\$7,200,000	\$500,000	\$-	\$10,268,212	\$600,000	\$426,821
J1089	\$4,258,719	\$-	\$-	\$-	\$3,263,069	\$11,000,000	\$500,000	\$-	\$18,521,788	\$680,000	\$1,172,179
J1090	\$-	\$-	\$-	\$-	\$-	\$5,500,000	\$500,000	\$-	\$5,500,000	\$360,000	\$190,000
J1103	\$-	\$-	\$-	\$-	\$-	\$100,000	\$-	\$-	\$100,000	\$80,000	\$-
J1117	\$-	\$-	\$-	\$-	\$-	\$1,100,000	\$500,000	\$-	\$1,100,000	\$320,000	\$-
J1119	\$6,000,000	\$-	\$-	\$-	\$-	\$6,500,000	\$500,000	\$-	\$12,500,000	\$2,400,000	\$-
J1172	\$-	\$-	\$-	\$-	\$-	\$6,868,622	\$500,000	\$-	\$6,868,622	\$200,000	\$486,862
J1173	\$870,000	\$-	\$-	\$-	\$-	\$11,500,000	\$500,000	\$-	\$12,370,000	\$320,000	\$917,000
J1178	\$-	\$-	\$-	\$-	\$-	\$750,000	\$500,000	\$-	\$750,000	\$260,000	\$-
Total	\$57,091,000	\$-	\$-	\$-	\$24,000,000	\$133,268,622	\$12,000,000	\$2,092,820	\$216,452,442	\$14,880,000	\$11,470,244



Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. Energy Resource Interconnection Service (ERIS) Network Upgrades and Network Resource Interconnection Service (NRIS) Network Upgrades are shown in Table 3. Shared Network Upgrades are shown in Table 4.

Table 3: ERIS & NRIS Upgrades (Planning level cost estimates)

Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
Vestaburg1-Vestaburg2 69 kV- Upgrade Vestaburg Bus tie conductor	WPSC	J1065, J978		\$60,000
Cut Majestic-Oneida 345 into Tompkins 345kV	METC	J1119		\$6,000,000
J875POI-Parr Rd 138kV- addition of the 12B7 breaker and associated switch	METC	J1173		\$370,000
J1089 Sub-Bell Rd 138kV-Reconductor	METC	J1089		\$210,000
J1089 SUB-Nornaj 138kV-Reconductor	METC	J1088, J1089		\$4,380,000
Donald-White lake 138kV- Rebuild	METC	J1071		\$19,420,000
Leoni-Parr Rd 138kV- Sag remediation A10 conditionality on MTEP 15938	METC	J1173		A10
Bell Rd- Claremont W 138kV- Sag Remediation A10 conditionality on MTEP 15836	METC	J1089		A10
Raisin-Whiting 138kV- Sag remediation	METC	J1173		\$500,000
Spaulding- PetisJ 138kV- Reconductor A10 conditionality on MTEP 15805	METC	J1043, J1064		A10
Hersey - Vestaburg 69kV to 138kV conversion	WPSC	J1065, J978		\$26,151,000
Cornell-Goss 138kV Sag remediation A10 conditionality on MTEP 15798	METC		J989	A10
Snckj-Vanat 138kV- Sag remediation plus A10 conditionality on MTEP 15852	METC		J1088, J1089	\$2,000,000 A10
Bennington- Snckj 138kV- Sag remediation	METC		J1088, J1089	\$2,000,000
Cornell-Bennington 138 kV- Sag remediation	METC		J1088, J1089	\$2,000,000
Redstone-Titbaw 138 kV- Rebuild the 13.5 Miles	METC		J1065	\$18,000,000



Table 4: Shared Network Upgrades (Planning level cost estimates)

Shared Network Upgrade	TO	Higher queued projects associated with SNU	Study projects associated with SNU	Cost of solution (\$)
Vestaburg 138-69 kV Transformer	WPSC	J589, J841	J978, J1065	\$3,093,500

Note:

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.
- 3) Analysis performed shows there are three study projects for Shared Network Upgrade cost allocation.
- 4) A detailed summary of the estimated cost of Network Upgrades on a per project basis can be found in Appendix B – Network Upgrade Table Per Project (CEII).

2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, stated that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to J978, J984, J989, J996, J1005, J1013, J1014, J1043, J1046, J1056, J1062, J1064, J1065, J1068, J1071, J1083, J1088, J1089, J1090, J1103, J1117, J1172, J1173, and J1178 projects included in the DPP 2018 April Michigan study cycle.

In this study, the power factor at the high-side of the generator substation for each inverter based project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that all projects meets the requirement to maintain 0.95 leading power factor, however, four projects do not meet the requirement to provide reactive power capability corresponding to 0.95 lagging power factor, as highlighted in red below in Table 5. Additional reactive support will be needed for these projects to meet the FERC requirement on reactive power capability prior to the completion of their GIA.

Table 5: FERC Order 827 Review Results

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J978	100	+/-32.8680	1x11.5	97.5	33.4	0.95	96.8	44.2	-0.91	Y	
J984	200	+/-65.7370	2x22	194.3	74.5	0.93	191.9	134.4	-0.82	Y	
J989	80	+/-26.2950	2x6	79.1	27.1	0.95	79	29.4	-0.94	Y	
J996	80	+/-26.2950	2x6	79.1	26.9	0.95	79	26.9	-0.95	Y	
J1005	200	+/- 65.7370	2x14	197.7	64.2	0.95	197.1	107.9	-0.88	Y	
J1013	80	+/-26.2950	2x6	79.4	36	0.91	79	28	-0.94	Y	
J1014	80	+/-26.2950	2x6	79.1	26.9	0.95	79	29.1	-0.94	Y	
J1043	374.4	+/-159.4940	NA	366.8	90.4	0.97	342.4	381	-0.67	N	30.161
J1046	50	+/-16.4340	NA	49.7	12.4	0.97	49.6	22.2	-0.91	N	3.936
J1056	200	+/-96.8640	2x6	195.8	77.8	0.93	194.3	142.6	-0.81	Y	
J1062	200	+/-96.8640	2x2	197.2	67.9	0.95	195.8	141.4	-0.81	Y	
J1064	200	+/-96.8640	2x6	196.1	80.7	0.92	193.9	145.7	-0.80	Y	
J1065	200	+/-96.8640	2x6	196.2	82	0.92	193.7	147.5	-0.80	Y	
J1068	200	+/-96.8640	2x6	195.7	80	0.93	192.9	152.9	-0.78	Y	
J1071	100	+/-32.8680	NA	98.7	22	0.98	98.4	50.4	-0.89	N	10.441
J1083	70	+/-23.0080	2x5	69.3	23.9	0.95	69.2	24.5	-0.94	Y	
J1088	150	+/-49.3030	NA	147.6	27.6	0.98	146.5	88	-0.86	N	20.914
J1089	170	+/-55.8760	NA	167.3	23.8	0.99	160.7	160.6	-0.71	N	31.189
J1090	90	+/-29.5820	NA	88.6	17.6	0.98	88.5	46.6	-0.88	N	11.521
J1103	20	+/-6.5740	1x3	19.9	9.1	0.91	19.9	7.3	-0.94	Y	
J1117	80	+/-26.2950	2x6	79.1	26.6	0.95	79	29.6	-0.94	Y	
J1119	600	NA	NA	NA	NA	NA	NA	NA	NA	Y	
J1172	50	+/-16.4340	4x4	49.6	28	0.87	49.6	24.1	-0.90	Y	
J1173	80	+/-26.2950	NA	78.7	8.3	0.99	78	53.2	-0.83	N	17.567

3. Model Development and Study Assumptions

3.1. Base Case Models

The origin of the DPP 2018 April Michigan models is the MTEP 18 models with the Bench Cases including all pre-queued projects and associated network upgrades, while the Study Cases contain all of the interconnection requests in DPP 2018 April Michigan Phase 1, in addition to all the facilities in the Bench Cases. Table 6 provides the list of different Scenarios study for DPP 2018 April Michigan.

- Bench Cases
 - 1)Bench-DPP_2018APR-MI_2023SPK_IN_LG
 - 2)Bench-DPP_2018APR -MI_2023SPK_Z_LG
 - 3)Bench-DPP_2018APR -MI_2023SH_85_Z_LG
 - 4)Bench-DPP_2018APR -MI_2023SH_70_IN_LP
 - 5)Bench-DPP_2018APR -MI_2023SH_85_IN_LG
- Study Cases
 - 1)Study-DPP_2018APR-MI_2023SPK_IN_LG
 - 2)Study-DPP_2018APR -MI_2023SPK_Z_LG
 - 3)Study-DPP_2018APR -MI_2023SH_85_Z_LG
 - 4)Study-DPP_2018APR -MI_2023SH_70_IN_LP
 - 5)Study-DPP_2018APR -MI_2023SH_85_IN_LG

Table 6: DPP 2018 April Michigan-Scenarios

Case Category	Case Name	Case Season	Flow on ITCT-IESO interface	Flow on METC-ATC DC line	Status of Ludington	Load Level
#1	DPP_2018APR-MI_2023SPK_IN_LG	Summer Peak 2023	1400 MW IESO to ITCT	5 MW ATC to METC	Generating (6 Gen)	Peak
#2	DPP_2018APR - MI_2023SPK_Z_LG	Summer Peak 2023	Zero Flow	5 MW ATC to METC	Generating (6 Gen)	Peak
#3	DPP_2018APR - MI_2023SH_85_Z_LG	Summer Shoulder 2023	Zero Flow	5 MW ATC to METC	Generating (5 Gen)	MTEP 18 Base SH (77-78%)
#4	DPP_2018APR - MI_2023SH_70_IN_LP	Summer Shoulder 2023	1600 MW IESO to ITCT	5 MW ATC to METC	Pumping (6 Pump)	70% Peak
#5	DPP_2018APR - MI_2023SH_85_IN_LG	Summer Shoulder 2023	1600 MW IESO to ITCT	5 MW ATC to METC	Generating (5 Gen)	MTEP 18 Base SH (77-78%)

3.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies

- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

3.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2018 April Michigan generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 33.10.0 and TARA version 1901_1.

3.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as Injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner FERC filed Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the Deliverability algorithm, to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

4. Thermal Analysis

The thermal analysis results for 2023 Summer Peak and Summer Shoulder scenarios show generator projects J978, J1043, J1064, J1065, J1065, J1071, J1088, J1089, J1119, and J1173 causing constraints. The details pertaining to the thermal analysis can be found in Appendix A – Cost Allocation Summary (CEII)

4.1. J978

The thermal analysis identified the generator to contribute to the following constraint:

1. Vestaburg1-Vestaburg2 69 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to upgrade Vestaburg Bus tie conductor. The cost estimate is \$0.06 million, with the project allocated 33% of the cost.

2. Mulliken-Portland 69kV, Mulliken- Chester 69kV, Vestaburg 138kV- to Vestaburg2 69kV, Altona-



Hinton 69kV, and Pierson-Hinton 69kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to convert Hersey-Vestaburg 69kV to 138kV. The cost estimate is \$26.151 million, with the project allocated 33% of the cost.

4.2. J1043

The thermal analysis identified the generator to contribute to the following constraint:

1. Spaulding- PetisJ 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to reconductor the line. MTEP Project #15805 will reduce the flow on this line below the required rating. If the MTEP project is not approved, there will be 41% of the cost assigned to J1043.

4.3. J1064

The thermal analysis identified the generator to contribute to the following constraint:

1. Spaulding- PetisJ 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to reconductor the line. MTEP Project #15805 will reduce the flow on this line below the required rating. If the MTEP project is not approved, there will be 59% of the cost assigned to J1043.

4.4. J1065

The thermal analysis identified the generator to contribute to the following constraint:

1. Vestaburg1-Vestaburg2 69 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to upgrade Vestaburg Bus tie conductor. The cost estimate is \$0.06 million, with the project allocated 67% of the cost.

2. Mulliken-Portland 69kV, Mulliken- Chester 69kV, Vestaburg 138kV- to Vestaburg2 69kV, Altona-Hinton 69kV, and Pierson-Hinton 69kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to convert Hersey-Vestaburg 69kV to 138kV. The cost estimate is \$26.151 million, with the project allocated 67% of the cost.

4.5. J1071

The thermal analysis identified the generator to contribute to the following constraint:

1. Donald-White lake 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild the line. The cost estimate is \$19.42 million, with the project allocated 100% of the cost.

4.6. J1088

The thermal analysis identified the generator to contribute to the following constraint:



1. J1089 SUB-Ncrnaj 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to reconductor the line. The cost estimate is \$4.38 million, with the project allocated 8% of the cost.

4.7. J1089

The thermal analysis identified the generator to contribute to the following constraint:

1. J1089 Sub-Bell Rd 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to reconductor the line. The cost estimate is \$0.21 million, with the project allocated 100% of the cost.

2. J1089 SUB-Ncrnaj 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to reconductor the line. The cost estimate is \$4.38 million, with the project allocated 92% of the cost.

3. Bell Rd- Claremont W 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to raise the sag limit on the line. MTEP Project #15836 will reduce the flow on this line below the required rating. If the MTEP project is not approved, there will be 100% of the cost assigned to J1089.

4.8. J1119

The thermal analysis identified the generator to contribute to the following constraint:

1. Tompkin-Tompkin1- Tompkin2 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to Cut Majestic - Oneida 345 into Tompkins 345kV. The cost estimate is \$6 million, with the project allocated 100% of the cost.

4.9. J1173

The thermal analysis identified the generator to contribute to the following constraint:

1. J875POI-Parr Rd 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is addition of the 12B7 breaker and associated switch. The cost estimate is \$0.37 million, with the project allocated 100% of the cost.

2. Leoni-Parr Rd 138kV- Sag remediation

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to perform a sag remediation. MTEP Project #15938 will reduce the flow on this line below the required rating. If the MTEP project is not approved, there will be 100% of the cost assigned to J1173.

3. Raisin-Whiting 138kV- Sag remediation



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to perform a sag remediation. The cost estimate is \$0.5 million, with the project allocated 100% of the cost.

5. Voltage Analysis

The voltage analysis results for 2023 Summer Peak and Summer Shoulder scenarios show that the study generators do not cause any voltage violations. The details pertaining to the voltage analysis can be found in [Appendix C – MISO ERIS Analysis \(CEII\)](#).

6. Deliverability Analysis

6.1. Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource (ER) project or to proceed with the system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

6.2. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator's output is not deliverable.

6.3. Deliverability Study Results

The limiting constraints (mon-con pairs) seen in the deliverability analysis for the 2023 Summer case are summarized in [Appendix D – MISO Deliverability Analysis \(CEII\)](#)

6.4. J978

This generator is found to be fully deliverable for 100 MW without any NRIS network upgrades.

6.5. J984

This generator is found to be fully deliverable for 200 MW without any NRIS network upgrades.

6.6. J989

This generator is determined to be fully deliverable for 80 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 9 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 7: NRIS Results for J989

J989 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)	9.44 MW (11.8%)
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Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERI Analysis?	Projects Associated with ERI Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Cornell-Goss 138kV	80	0.0583	No		J989	0%	A10 Conditionality on MTEP19 Target A Project #15798

6.7. J996

This generator is found to be fully deliverable for 80 MW without any NRIS network upgrades.

6.8. J1005

This generator is found to be fully deliverable for 200 MW without any NRIS network upgrades.

6.9. J1013

This generator is found to be fully deliverable for 80 MW without any NRIS network upgrades.

6.10. J1014

This generator is found to be fully deliverable for 80 MW without any NRIS network upgrades.

6.11. J1043

This generator is found to be fully deliverable for 374.4 MW without any NRIS network upgrades.

6.12. J1046

This generator is found to be fully deliverable for 50 MW without any NRIS network upgrades.

6.13. J1056

This generator is found to be fully deliverable for 200 MW without any NRIS network upgrades.

6.14. J1062

This generator is found to be fully deliverable for 200 MW without any NRIS network upgrades.

6.15. J1064

This generator is found to be fully deliverable for 200 MW without any NRIS network upgrades.

6.16. J1065

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 8 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 8: NRIS Results for J1065

J1065 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERI and case assumptions)				110.51 MW (55.3%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERI Analysis?	Projects Associated with ERI Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Redstone-Titbaw 138 kV	200	0.4947	No		J1065	100%	\$18,000,000

6.17. J1068

This generator is found to be fully deliverable for 200 MW without any NRIS network upgrades.

6.18. J1071

This generator is found to be fully deliverable for 100 MW without any NRIS network upgrades.

6.19. J1083

This generator is found to be fully deliverable for 70 MW without any NRIS network upgrades.

6.20. J1088

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 9 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 9: NRIS Results for J1088

J1088 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIIS and case assumptions)				14.9 MW (9.9%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Snckj-Vanat 138kV	38.65	0.2078	No		J1088, J1089	45.6%	\$2,000,000 Plus A10 conditionality on MTEP19 Project #15852
Bennington- Snckj 138kV	102.5	0.2078	No		J1088, J1089	45.6%	\$2,000,000
Cornell-Bennington 138 kV	150	0.2078	No		J1088, J1089	45.6%	\$2,000,000

6.21. J1089

This generator is determined to be fully deliverable for 170 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 10 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 10: NRIS Results for J1089

J1089 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIIS and case assumptions)				16.88 MW (9.9%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Snckj-Vanat 138kV	43.8	0.2186	No		J1088, J1089	54.4%	\$2,000,000 Plus A10 conditionality on MTEP19 Project #15852
Bennington- Snckj 138kV	116.17	0.2186	No		J1088, J1089	54.4%	\$2,000,000
Cornell-Bennington 138 kV	170	0.2186	No		J1088, J1089	54.4%	\$2,000,000

6.22. J1090

This generator is found to be fully deliverable for 90 MW without any NRIS network upgrades.

6.23. J1103

This generator is found to be fully deliverable for 20 MW without any NRIS network upgrades.

6.24. J1117

This generator is found to be fully deliverable for 80 MW without any NRIS network upgrades.

6.25. J1119

This generator is found to be fully deliverable for 600 MW without any NRIS network upgrades.

6.26. J1172

This generator is found to be fully deliverable for 50 MW without any NRIS network upgrades.

6.27. J1173

This generator is found to be fully deliverable for 80 MW without any NRIS network upgrades

6.28. J1178

This generator is found to be fully deliverable for 65 MW without any NRIS network upgrades.

7. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) Analysis tests for Network Upgrades driven by higher queued interconnection projects was performed for this System Impact Study.

The maximum MW impacts and Shared Network Upgrade (SNU) cost allocations appear in Table 11.

Table 11: Maximum MW Impact and SNU Cost Allocations

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility (\$)
Vestaburg 138-69 kV Transformer	DPP 2016 Aug MI	J589	1.7	\$ 3,093,500	\$153,330
	DPP 2017 Aug MI	J841	9.5		\$847,350
	DPP 2018 Apr MI	J1065	15.3		\$1,372,797
	DPP 2018 Apr MI	J978	8.0		\$720,023

8. Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

8.1. Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade identified in System Impact Study was provided by the corresponding transmission owning company.

8.2. Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

Cost Allocation Methodology for Thermal Constraints

1. With all Study Group generation projects dispatched in the Post Case, all thermal constraints are

identified.

2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any network upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$\text{Cost of NU} = \frac{\text{Project A cost portion of NU} \times \text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contrution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU is required.



Appendix A – Cost Allocation Summary (CEII)

Appendix B – Network Upgrade Table Per Project (CEII)

Appendix C – MISO ERIS Analysis (CEII)

Appendix D – MISO Deliverability Analysis (CEII)